Assignment

ICT607 Artificial Intelligence for Cyber Security

Due Date: As stated in the LMS

Individual Task: Every student must submit the Assignment via LMS by the due date.

Late Penalty: 10% per day penalty for delayed submissions unless prior extension of deadline is obtained from the unit coordinator. Work submitted more than 10 days late will not be marked. This is to enable students to receive feedback within a reasonable amount of time.

Description of the assessment:

In this assignment, you will be required to detect anomaly(s) in a dataset using any machine learning technique.

Some of the guidelines for the task:

- **Dataset**: Find a dataset related to anomaly in regards to cyber security. If you are not sure about the relevance of your dataset then ask the tutor. If you failed to find a suitable dataset ask the tutor who might help you find a dataset for you.
- **Pre-processing**: Acquired dataset might not be ready to be used in a ML algorithm. Make sure you perform necessary pre-processing on the dataset.
- **ML algorithm**: You have to decide which ML algorithm you would use for your task based on (i) applicability in the chosen dataset/context (ii) available library or implementations etc.
- **Programming environment**: It is recommended that you use Python programming language using any suitable IDE you prefer.
- **Visualization**: You can use any method of visualization to present the outcome.

Deliverables and rubrics:

Complete all the following components and zip them in a folder and upload the folder in the LMS using the appropriate link.

- **Report [85 marks]:** Submit a word/pdf document containing following minimal items:
    - **The cover page:** the first page of the report should contain the assignment name, your personal details etc.
    - **Executive summary** [10 marks]: Briefly explain what is done in this assignment.

- **Dataset familiarization** [10 marks]: Discuss about the dataset in detail, such as - what the dataset was about, when was it collected, how many records, what are the features, some sample snapshots etc.
- **Pre-processing** [15 marks]: Demonstrate what kind of pre-processing was needed with justifications. How you performed those pre-processing.
- **ML algorithm** [15 marks]: Describe which algorithm you have chosen with justification. Discuss which library or implementation have you used in your ML analysis.
- **Accuracy** [10 marks]: Discuss how was the accuracy of the algorithm in your analysis
- **Visualization and conclusion** [10 marks]: Finally present your findings from your analysis.
- **Code explanation**[15 marks]: Briefly explain how your code should be executed. This should include a brief explanation of all necessary files and dependencies, location of the dataset, necessary library installations etc.

- **Code [15 marks]**:
  - Submit the code and necessary dependency files. Make sure you organize your code in such a way that the marker can execute your code smoothly.

Note: For any further clarification or feedback please feel free to contact the teaching staff.